Akoizo Original Practice - 2027 Rotation Form 3

Division C | Circuit Lab | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) An LED’s forward voltage is…

Unit: Switches, relays, diodes, and LEDs.
Worked-case data: For Vs = 5 V, Vf = 2 V, and R = 300 Ω, estimate current.

A. Always zero

B. Equal to any source

C. Device- and condition-dependent

D. Always 0.7 V

Answer: _____

2. (2 points) A watt-hour measures…

Unit: Current, voltage, resistance, power, and energy.
Worked-case data: A 6 V source drives a 300 Ω resistor for 60 s. Find current, power, and energy.

A. Resistance

B. Charge alone

C. Energy

D. Current

Answer: _____

3. (2 points) Negative solved current means…

Unit: Kirchhoff laws and network problem solving.
Worked-case data: At a node, 12 mA and 8 mA enter. One branch leaves. Find its current.

A. No voltage exists

B. Opposite the assumed direction

C. Impossible circuit necessarily

D. Negative charge disappears

Answer: _____

4. (2 points) A diode is generally…

Unit: Switches, relays, diodes, and LEDs.
Worked-case data: For Vs = 5 V, Vf = 2 V, and R = 300 Ω, estimate current.

A. A mechanical switch

B. Nonlinear

C. An ideal fixed resistor

D. Always conducting both ways

Answer: _____

5. (4 points) Why is charge not consumed?

Unit: Current, voltage, resistance, power, and energy.
Worked-case data: A 6 V source drives a 300 Ω resistor for 60 s. Find current, power, and energy.

Answer: _____
_____
_____

6. (4 points) Write a source-two-resistor loop equation.

Unit: Kirchhoff laws and network problem solving.
Worked-case data: At a node, 12 mA and 8 mA enter. One branch leaves. Find its current.

Answer: _____
_____
_____

7. (4 points) Name two LED faults.

Unit: Switches, relays, diodes, and LEDs.
Worked-case data: For Vs = 5 V, Vf = 2 V, and R = 300 Ω, estimate current.

Answer: _____
_____
_____

8. (4 points) Why is Ah not sufficient for battery energy?

Unit: Current, voltage, resistance, power, and energy.
Worked-case data: A 6 V source drives a 300 Ω resistor for 60 s. Find current, power, and energy.

Answer: _____
_____
_____

9. (4 points) Explain a nodal branch current.

Unit: Kirchhoff laws and network problem solving.
Worked-case data: At a node, 12 mA and 8 mA enter. One branch leaves. Find its current.

Answer: _____
_____
_____

10. (4 points) Why not infer brightness from current alone?

Unit: Switches, relays, diodes, and LEDs.
Worked-case data: For Vs = 5 V, Vf = 2 V, and R = 300 Ω, estimate current.

Answer: _____
_____
_____

11. (2 points) A 9 V ideal battery drives a 4.5 ohm resistor. Find current.

Answer: _____
_____
_____

12. (2 points) A series pair has resistances 5 and 7 ohms. Find equivalent resistance.

Answer: _____
_____
_____

13. (2 points) Two identical 10 ohm resistors are placed in parallel. Find equivalent resistance.

Answer: _____
_____
_____

14. (2 points) An ideal resistor carries 2 A at 9 V. Find power.

Answer: _____
_____
_____

15. (2 points) A node has incoming currents 6 A and 2 A and one outgoing current. Find the outgoing current.

Answer: _____
_____
_____

16. (2 points) Explain why an ideal voltmeter is connected in parallel rather than used as a series current path.

Answer: _____
_____
_____

17. (2 points) Give the output of an AND gate for inputs 1 and 0, and explain.

Answer: _____
_____
_____

18. (2 points) A silicon diode is reverse biased below breakdown. Contrast its ideal current with a forward-biased conducting diode.

Answer: _____
_____
_____

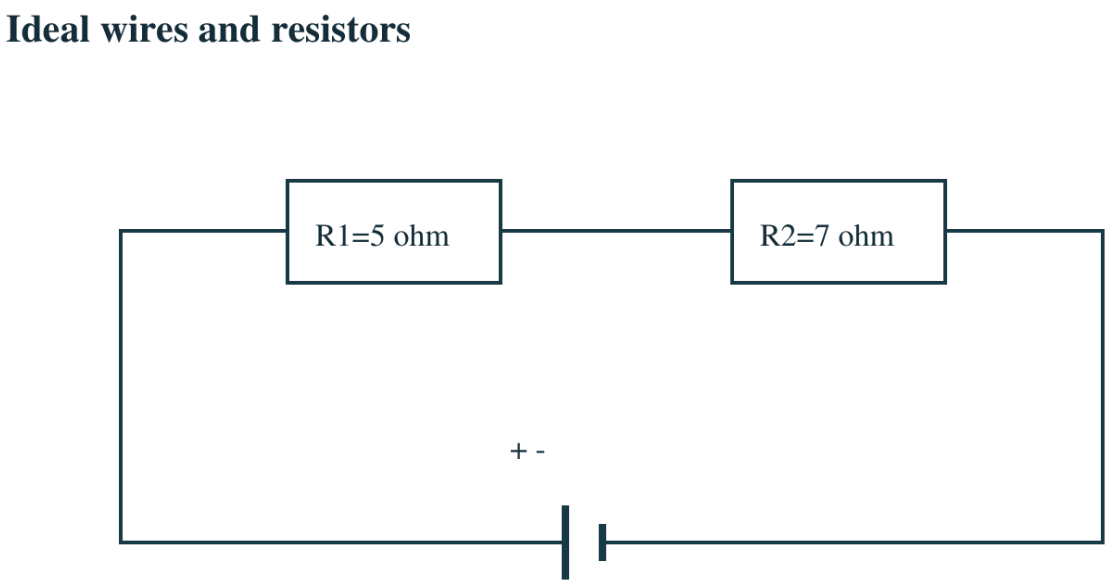
19. (2 points) A closed DC loop has a 9 V source and a 4 V resistor drop. Find the remaining series resistor drop and state the governing law.

Answer: _____
_____
_____

20. (2 points) Two identical LEDs must be equally bright in a low-voltage teaching circuit. Explain why their branch currents and current-limiting arrangements matter.

Answer: _____
_____
_____

21. (2 points) Use Figure F1. Calculate equivalent resistance.



F1. Series resistor circuit (original synthetic teaching figure)

Answer: _____
_____
_____

22. (2 points) Use Figure F1. Calculate loop current.

Refer to F1. Series resistor circuit (original synthetic teaching figure) shown with the first question in this figure group.

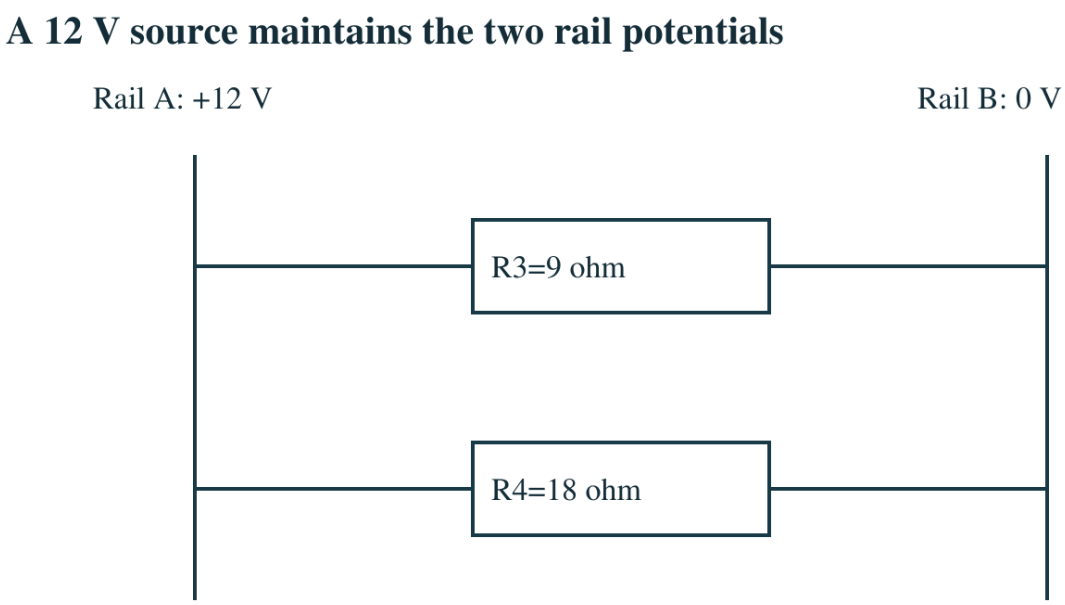
Answer: _____
_____
_____

23. (2 points) Use Figure F1. Calculate the potential difference across R1.

Refer to F1. Series resistor circuit (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

24. (2 points) Use Figure F2. Find current through R3.



F2. Two parallel branches (original synthetic teaching figure)

Answer: _____
_____
_____

25. (2 points) Use Figure F2. Find current through R4.

Refer to F2. Two parallel branches (original synthetic teaching figure) shown with the first question in this figure group.

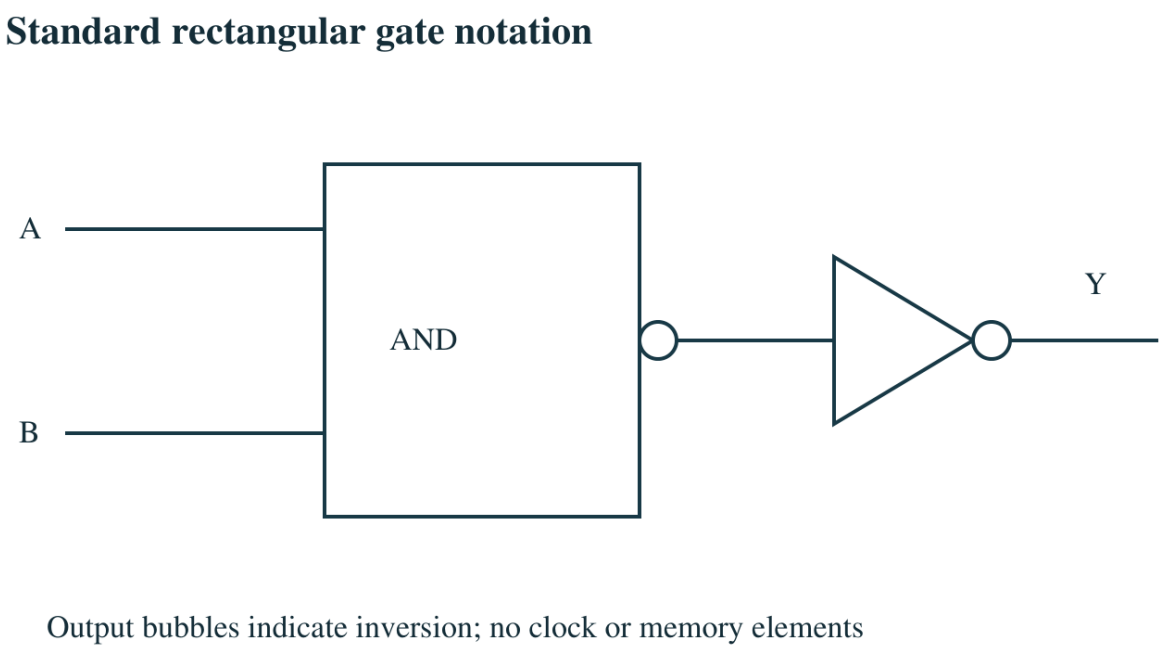
Answer: _____
_____
_____

26. (2 points) Use Figure F2. Find equivalent resistance between the rails.

Refer to F2. Two parallel branches (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

27. (2 points) Use Figure F3. Find Y for A=1 and B=0.



F3. Combinational logic (original synthetic teaching figure)

Answer: _____
_____
_____

28. (2 points) Use Figure F3. Find Y for A=1 and B=1.

Refer to F3. Combinational logic (original synthetic teaching figure) shown with the first question in this figure group.

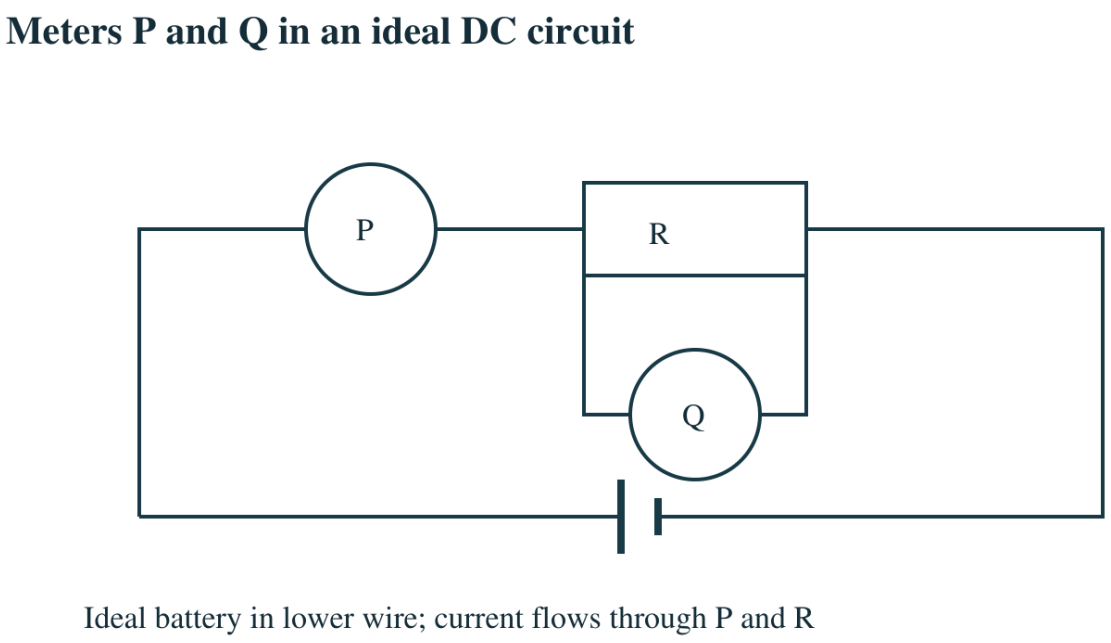
Answer: _____
_____
_____

29. (2 points) Use Figure F3. Name the single gate equivalent to the complete diagram and justify.

Refer to F3. Combinational logic (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

30. (2 points) Use Figure F4. Which meter position is appropriate for a voltmeter? Explain using the wiring.



F4. Measurement connections (original synthetic teaching figure)

Answer: _____
_____
_____

31. (2 points) Use Figure F4. Which meter position is appropriate for an ammeter? Explain.

Refer to F4. Measurement connections (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

32. (2 points) Use Figure F4. Describe the desired input resistance of the meter at Q and its purpose.

Refer to F4. Measurement connections (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____